

Recursive Spectral Fibers on Simplicial Cobordisms

A geometric program for quarks, color, fermion statistics, Fock space, and baryons

Tessera - cobordism programme

Implementation is tracked by GitHub epic [#763](#).

Abstract

This paper proposes a single geometric formulation for the particle content already suggested by Tessera’s cobordism experiments. A connected, high-modularity simplicial component is treated as one effective vertex at the next resolution. The component’s selected localized Hodge eigenspace is its fiber, and the couplings between components induce transport between those fibers. Repeating this operation produces a nested, potentially fractal hierarchy of complexes. No independent gauge field, particle label, or auxiliary lattice is introduced. Each edge supplies one two-level occupation mode and complex geometric data, while the generally entangled state lives on the exterior Fock space of all active edge modes. Connections, exchange signs, and observables are derived from that state and the Hodge/Regge operators already present in the construction.

The proposal has a substantial exact core. Static Schur reduction proves when a component may be replaced by a response vertex without changing supported boundary quadratic energies. Nonzero spectral bands instead use the energy-dependent Feshbach–Schur map, or a certified Craig–Bampton/AMLS linear surrogate. Simplicial gluing acts on the one-particle chain space; fermionic second quantization then turns direct sums into graded tensor products and coupling blocks into hopping terms. Three oriented edge-mode factors form the exact exterior algebra $\Lambda^\bullet \mathbb{C}^3 = \mathbf{1} \oplus \mathbf{3} \oplus \bar{\mathbf{3}} \oplus \mathbf{1}$; the one-occupation sector is a color qutrit, its bilinears close $\mathfrak{su}(3)$, the three-occupation sector is a color singlet, and the grading gives the fermionic exchange sign and canonical anticommutation relations. A rank- r connection is derived from overlap of neighboring spectral frames. At rank three its determinant line and projective $SU(3)/\mathbb{Z}_3$ transport are retained rather than choosing an unrecorded cube-root branch. Closed holonomies are gauge-invariant observables rather than new degrees of freedom. Successive cobordism interactions generate the finite stages of an inductive-limit Fock space.

The physical identification remains a hypothesis to be tested. A quark is proposed to be a persistent, odd-parity, rank-three spectral fiber localized on a high-modularity component; a proton is three such components bound into one persistent supercomponent, occupying a normalized color wedge, carrying baryon number $+1$, electric charge $+1$, and a total-space spin- $1/2$ holonomy. The paper separates exact identities, results already supported by Tessera, and new falsifiable conjectures. It also states an implementation program in which analytic or structure-exact operations are used before iterative numerical work, so the model can scale without turning its defining observables into uncontrolled approximations.

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1 Epistemic status and design constraint

Three kinds of statement are deliberately distinguished:

1. **Exact identity** — follows algebraically from the stated finite complex, orientation, and inner product.
2. **Existing Tessera evidence** — measured by an existing experiment, with the scope and residual reported in the repository.
3. **Proposed physical identification** — a new hypothesis with an explicit falsification test.

The governing constraint is parsimony. The ontology is limited to:

- an oriented simplicial complex and its cobordisms;
- a complex squared length and one two-level occupation mode on each edge;
- incidence, Hodge, and Regge operators derived from that data;
- a generally entangled boundary/Fock state on those modes; and
- simplicial gluing followed by fermionic second quantization.

Spectral fibers, color frames, connections, Wilson loops, particle sectors, and coarse vertices are *derived views* of that same data. They are not separately sampled fields. This is important both scientifically and computationally: adding a new independent field could fit a desired answer, while deriving every readout from one complex leaves the construction falsifiable.

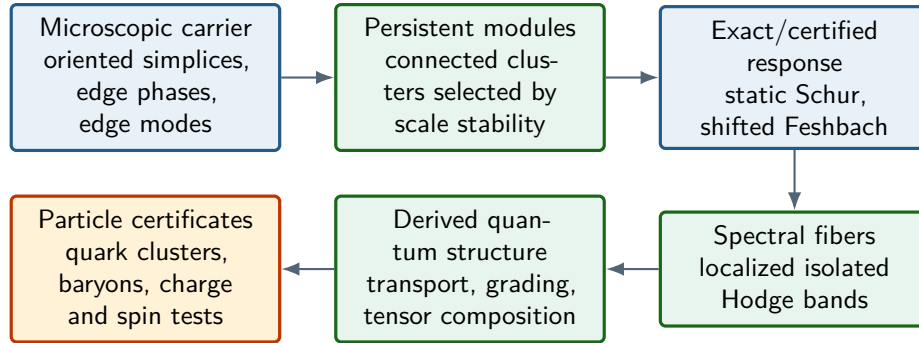


Figure 1: Concept map for the recursive complex construction. Colors encode epistemic status, not physical sectors: blue is established or exact machinery, green is a derived observable, and amber is a proposed physical identification.

2 Present evidence in Tessera

Two existing results motivate the construction.

First, the state-operation-cobordism experiments show that the Hodge-carried register is an isometry to machine precision and that its spectral value reproduces the quantum transition amplitude for every operation that the tested geometry actually carries. Generic fixed-complexity operations can remain obstructed, and the obstruction is visible both as a residual floor and as leakage from the carried subspace. The claim is therefore not that every finite complex realizes every gate; it is that a realized, isometrically embedded register computes the corresponding amplitude. See [cobordism-results.md](#).

Second, interaction-history complexes exhibit a stable near-four-dimensional spectral regime. The

strongest reported measurements approach, but do not yet prove, an exact spectral dimension of four. The current status and finite-size caveats are recorded in `h_ds4_status.md`. Diffusion-based spectral dimension on simplicial quantum geometries has important precedent in causal dynamical triangulations [1]; the Tessera evidence is an independent result for a different construction and should be compared at the level of the return-probability estimator and its finite-size window.

The current proton animation adds a third, more preliminary observation. It starts with the phase pattern $\{1, \omega, \omega^2\}$ and evaluates its singlet diagnostics while a joint Regge-Hodge stationarity objective changes the complex. It no longer forces register holes to appear, and holes have not re-emerged in the current construction. That negative result is useful: the proposed quark should therefore not be defined as a hole. It will be sought as a persistent modular spectral cluster, while Betti numbers remain independent topological observables.

3 The microscopic geometric state

Let K be a finite oriented simplicial complex. For every edge e , store the complex squared length

$$z_e = \rho_e e^{i\theta_e} \in \mathbb{C}$$

and attach the two-level occupation factor $\mathcal{H}_e = \text{span}\{|0\rangle_e, |1\rangle_e\}$. The phase of z_e is not a second link field; it is part of the existing complex edge geometry. Saying that an edge “carries a qubit” means that it carries this local mode algebra, not that the global state is forced to be a product of normalized vectors q_e .

Writing $\mathfrak{h}_K = \text{span}\{|e\rangle : e \in K_1\}$ for the one-particle edge space, the microscopic quantum carrier is

$$\mathcal{H}_K = \mathcal{F}_-(\mathfrak{h}_K) = \Lambda^\bullet \mathfrak{h}_K \cong \widehat{\bigotimes_{e \in K_1} \mathcal{H}_e},$$

and a boundary state is a vector or density operator on $\mathcal{H}_K$. It may be entangled. A one-particle color state $a_\phi^\dagger |0\rangle$ and the nonseparable proton-spin sectors are therefore native states, not exceptions to the ontology. For an isolated occupied band with projector P , the corresponding quasi-free reference state has covariance

$$\Gamma_{ef} = \langle a_f^\dagger a_e \rangle = P_{ef}, \quad \langle n_e \rangle = P_{ee}.$$

Thus a per-edge Bloch vector or occupation is a derived marginal/readout. The quasi-free state is a useful analytic baseline, not a restriction: finitely many interacting non-Gaussian sectors are represented explicitly by the lazy Fock construction of Section 11.

Let

$$\partial_k : C_k(K) \longrightarrow C_{k-1}(K), \quad \partial_{k-1} \partial_k = 0$$

be the oriented boundary maps and let $\mathbf{W}_k(\mathbf{z})$ be the metric weight on $\mathbf{k}$ -chains. With the weighted adjoint

$$\partial_k^* = W_k^{-1} \partial_k^\dagger W_{k-1},$$

the degree- k Hodge Laplacian is

$$L_k = \partial_{k+1} \partial_{k+1}^* + \partial_k^* \partial_k.$$

In the positive metric regime this is self-adjoint in the W_k inner product. In the signed Lorentzian regime it can be non-normal; then left and right spectral frames and their biorthogonal condition numbers must be reported rather than silently treating L_k as Hermitian.

The geometry evolves toward joint stationary points of the existing Regge and Hodge functionals. In emergence mode, particle-specific observables below are read after optimization and are not inserted as target terms. Controlled synthesis mode may pin a carrier to test realizability, but that is a separate experiment.

This operator stack sits on established foundations: Regge calculus encodes piecewise-flat gravity in simplicial deficit angles [2], discrete exterior calculus supplies metric-dependent chain/cochain operators [3], and combinatorial Hodge spectra on simplicial complexes have a developed spectral theory [4]. Tessera's proposal is not a replacement for those constructions; it is a constrained use of them as the sole source of the later particle readouts.

4 A component is an exact static response vertex

Partition the k -cells of a connected component into interface cells B and interior cells I , and block its Hodge operator as

$$L = \begin{pmatrix} L_{BB} & L_{BI} \\ L_{IB} & L_{II} \end{pmatrix}.$$

In the positive self-adjoint regime, after projecting out incompatible interior zero modes, minimization over the interior has the exact solution

$$x_I^* = -L_{II}^+ L_{IB} x_B,$$

and the exact effective boundary operator

$$\boxed{L_{\text{eff}} = L_{BB} - L_{BI} L_{II}^+ L_{IB}}.$$

Here $+$ denotes the Moore-Penrose inverse on the supported interior subspace. For every compatible boundary value,

$$\min_{x_I} \begin{pmatrix} x_B \\ x_I \end{pmatrix}^\dagger L \begin{pmatrix} x_B \\ x_I \end{pmatrix} = x_B^\dagger L_{\text{eff}} x_B.$$

This is the precise static, or zero-frequency, sense in which a connected component can be replaced by a coarse response vertex. In a Hermitian indefinite regime the same equation is a stationarity condition, not a minimum. For a non-normal block it is simply block elimination; solvability requires

$$L_{IB}x_B \perp \ker L_{II}^\dagger.$$

The plain Schur complement does *not* preserve the nonzero spectrum. For a spectral parameter λ such that $L_{II} - \lambda I$ is invertible, define the exact Feshbach–Schur response

$$F_B(\lambda) = L_{BB} - \lambda I - L_{BI}(L_{II} - \lambda I)^{-1}L_{IB}.$$

Then

$$\lambda \in \text{spec } L \iff 0 \in \text{spec } F_B(\lambda),$$

with algebraic multiplicities in the supported finite-dimensional sector. At an interior resonance the inverse is replaced only after checking the compatibility condition $L_{IB}x_B \perp \ker(L_{II} - \lambda I)^\dagger$ and retaining the resonant interior modes explicitly. Thus harmonic response uses $F_B(0)$, while a localized band centered at λ_C uses $F_B(\lambda)$ over a stated frequency window. A linear reduced eigenproblem may instead retain interface constraint modes plus selected fixed-interface modes using Craig–Bampton component-mode synthesis or AMLS; that route is certified approximation whose error is controlled by residuals and separation from discarded modes, not an exact spectral identity [5–7].

The effective blocks between coarse components become operator-valued links. A harmonic or retained interior mode is not discarded; it becomes an explicit stalk/fiber coordinate attached to the response vertex.

For graph Laplacians this is the classical Kron reduction by Schur complement [8]. Spectral graph reduction provides related approximation guarantees when additional coarsening or truncation is performed [9]. The extension proposed here is to apply static response reduction degree by degree to weighted Hodge blocks, and shifted Feshbach or certified component-mode reduction to nonzero bands, while retaining localized zero, resonant, and selected interior modes as explicit fiber coordinates.

5 Recursive spectral fibers

Let $P_\ell = \{C_v^\ell\}$ be an intrinsic partition at scale ℓ into persistent connected components. At $\ell = 0$ the object is the microscopic simplicial complex K_0 . After the first elimination the honest coarse object is generally not another simplicial complex: it is an operator-valued response network $\mathcal{R}_{\ell+1}$ whose vertices carry vector spaces and whose links carry linear response blocks. A cellular sheaf on the quotient graph is a natural realization when the blocks admit compatible restriction-map factorization [10]; otherwise Tessera retains the more general response network and does not invent incidence maps that the reduction did not determine.

Within component $\mathbf{C}$, choose an isolated localized spectral band and, in the positive self-adjoint regime, a weighted orthonormal frame

$$\Phi_C = (\phi_1, \dots, \phi_r), \quad \Phi_C^\dagger W_C \Phi_C = I_r.$$

The derived fiber is

$$E_C = \text{Ran } \Phi_C.$$

In a Hermitian indefinite regime record the inertia of $\Phi_C^\dagger W_C \Phi_C$ and normalize it to a signature matrix $J_C = \text{diag}(I_p, -I_q)$. Negative Krein signature is a certificate, not an automatic identification with an antiparticle. In a non-normal regime use matched right and left frames Φ_C, Ψ_C with $\Psi_C^\dagger W_C \Phi_C = I$ and report both residuals and the frame condition number.

It need not be a harmonic space and therefore need not be supported by a hole. What it does require is a spectral gap, localization, and persistence. A candidate component is accepted only if all of the following remain stable across a stated range of scales:

- high modularity and low conductance relative to neighboring cuts;
- a localized spectral projector with stable rank;
- a nonzero band gap separating it from discarded modes;
- overlap with its predecessor and successor components;
- lifetime across multiple cobordism frames; and
- small external transport leakage.

This gives a type-stable hierarchy of response objects

$$\cdots \longrightarrow \mathcal{R}_2 \longrightarrow \mathcal{R}_1 \longrightarrow K_0$$

in which a response vertex at one level resolves into a connected microscopic component plus retained stalk coordinates at the next finer level. “Self-similar” refers to closure of the response-network data type, not to a claim that every reduced operator is a simplicial Hodge Laplacian. A fractal-like pattern is permitted but not required: measured scaling of module count, volume, boundary size, and spectral gap decides whether the hierarchy is statistically self-similar.

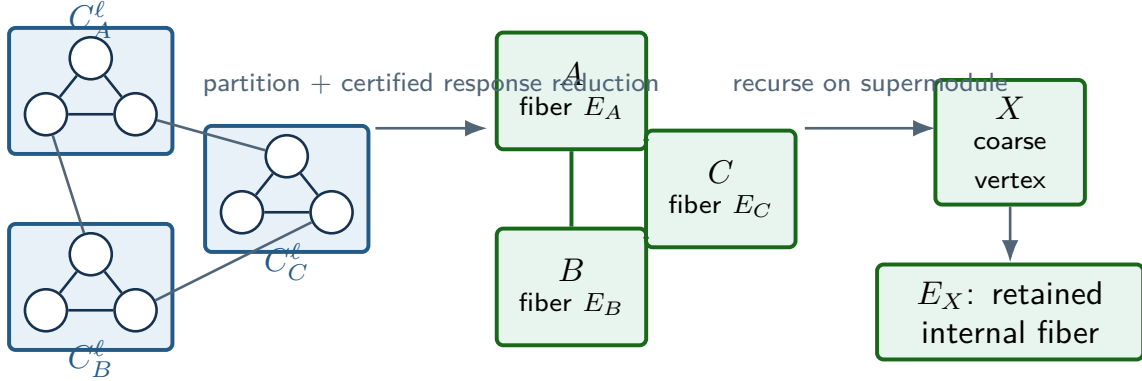
Community objectives supply deterministic cluster candidates [11], while network renormalization supplies tests for genuine self-similarity rather than visual resemblance [12]. The partition is therefore a measured part of the analysis: a recursively drawn pattern is not evidence of a fractal unless its scaling observables survive a refinement window. The current `ModularityOptimizer` uses Newman–Girvan modularity on a combinatorial one-skeleton; it is a heuristic proposal generator that does not see signed or complex Hodge weights and is subject to the modularity resolution limit [13]. Every accepted fiber is therefore conditioned on independent, weight-aware gap, localization, leakage, persistence, and refinement certificates.

6 Interactions and the expanding Hilbert space

Two operations must not be conflated. For the Cartesian product of chain complexes A and B , the graded tensor differential is the exact rule

$$d_{A \widehat{\otimes} B}(a \otimes b) = d_A a \otimes b + (-1)^{\deg a} a \otimes d_B b.$$

$K_0 / \mathcal{R}_\ell$: **connected modules** $\mathcal{R}_{\ell+1}$: **response network** $\mathcal{R}_{\ell+2}$: **recurse**



Static response is exact; band response carries a window and residual.

Figure 2: One recursive step. Persistent connected modules become stalk-bearing vertices of an operator-valued response network. Static response is preserved by the supported Schur complement; nonzero bands use shifted Feshbach or certified component-mode reduction. Selected internal modes remain attached as fibers, and a persistent supermodule can be reduced again at the next scale.

For a noninteracting product with product metric,

$$L_{A \hat{\otimes} B} = L_A \otimes I + I \otimes L_B,$$

so one-particle eigenvalues add and eigenvectors tensor. This identity is about a product complex, not about gluing two cobordisms.

Actual simplicial gluing is a pushout along a shared boundary. At the one-particle level it produces a chain space assembled from direct sums modulo boundary identifications (equivalently described by the relevant Mayer–Vietoris sequence) and a block operator

$$L_{A \cup B} = \begin{pmatrix} L_A & C_{AB} \\ C_{BA} & L_B \end{pmatrix}$$

in a basis adapted to the two interiors. The coupling blocks are induced by the connecting simplices and shared-boundary constraints; they are not a Kronecker interaction term.

The expanding Hilbert space follows after applying the fermionic Fock functor to the one-particle space $\mathfrak{h}$. The exact identities are

$$\mathcal{F}_-(\mathfrak{h}_A \oplus \mathfrak{h}_B) \cong \mathcal{F}_-(\mathfrak{h}_A) \hat{\otimes} \mathcal{F}_-(\mathfrak{h}_B),$$

and

$$d\Gamma(L_A \oplus L_B) = d\Gamma(L_A) \hat{\otimes} I + I \hat{\otimes} d\Gamma(L_B).$$

For the coupling block,

$$d\Gamma(C_{AB} + C_{BA}) = \sum_{ij} (C_{AB})_{ij} a_{A,i}^\dagger a_{B,j} + \text{h.c.},$$

so geometric connections become hopping terms without adding a new field. If the one-particle eigenvalues are $\lambda_1, \dots, \lambda_M$, then the free many-body spectrum is the set of occupation subset sums $\sum_i n_i \lambda_i$, $n_i \in \{0, 1\}$, rather than the one-particle pairwise spectrum being relabeled as a Fock spectrum [14].

At the selected-fiber level, an interaction grows the carried space as

$$\mathcal{H}_{AB} = E_A \hat{\otimes} E_B,$$

and a later interaction appends another factor. This is a statement about state-space composition after second quantization, not the topology of the glued chain complex. If J_C embeds an abstract state into the geometric carrier, exact amplitude preservation requires

$$J_C^\dagger W_C J_C = I.$$

Tensor products preserve isometry exactly. If $G = J_C^\dagger W_C J_C$ has Gram defect $\varepsilon = \|G - I\|$, then

$$|a^\dagger G b - a^\dagger b| \leq \varepsilon \|a\| \|b\|,$$

and two tensor factors obey

$$\varepsilon_{AB} \leq \varepsilon_A + \varepsilon_B + \varepsilon_A \varepsilon_B.$$

Thus the amplitude claim has an explicit, composable error budget.

Cobordism composition as a map between boundary state spaces is the organizing idea of topological field theory [15]; the general-boundary program makes the region/boundary assignment explicit for quantum theory [16], and categorical quantum mechanics formalizes tensor composition and diagrammatic process semantics [17]. Tessera keeps only the parts that can be realized by its finite simplicial carrier and tests the resulting map numerically rather than assuming topological invariance.

7 A triangle carries the exact color algebra

Consider the three edge-mode factors around an oriented triangle and interpret $|1\rangle$ as an occupied edge mode. Choosing an oriented ordering (e_1, e_2, e_3) identifies their graded tensor product with the exterior algebra

$$(\mathbb{C}^2)^{\hat{\otimes} 3} \cong \Lambda^\bullet \mathbb{C}^3 = \mathbf{1} \oplus \mathbf{3} \oplus \bar{\mathbf{3}} \oplus \mathbf{1}.$$

The orientation of one triangle fixes the ordering up to a cyclic, hence even, permutation, so the

local wedge sign is unambiguous. Globally the exterior algebra $\Lambda^\bullet \mathfrak{h}_K$ and the CAR are intrinsic; only a compilation into tensor-product qubits or bitsets needs a deterministic mode order and the corresponding permutation parity. A Kasteleyn orientation is useful for two-dimensional surface-dimer Pfaffians but is not required to define this abstract Fock space [18]. A genuine continuum spinor interpretation is a separate question addressed by the rotation certificate below.

The sectors have occupation number $N = 0, 1, 2, 3$:

Sector	Basis dimension	Color interpretation	Fermion parity
$\Lambda^0 \mathbb{C}^3$	1	vacuum	even
$\Lambda^1 \mathbb{C}^3$	3	fundamental color triplet	odd
$\Lambda^2 \mathbb{C}^3$	3	antisymmetric anti-triplet	even
$\Lambda^3 \mathbb{C}^3$	1	color singlet	odd

Let $a_i^\dagger, a_i$ be the exterior creation and contraction operators. They satisfy the canonical anticommutation relations exactly:

$$\{a_i, a_j\} = 0, \quad \{a_i^\dagger, a_j^\dagger\} = 0, \quad \{a_i, a_j^\dagger\} = \delta_{ij}.$$

On the one-occupation sector, the bilinears

$$E_{ij} = a_i^\dagger a_j$$

satisfy

$$[E_{ij}, E_{k\ell}] = \delta_{jk} E_{i\ell} - \delta_{i\ell} E_{kj}.$$

The six Hermitian off-diagonal combinations together with

$$H_1 = E_{11} - E_{22}, \quad H_2 = \frac{E_{11} + E_{22} - 2E_{33}}{\sqrt{3}}$$

are the eight generators of $\mathfrak{su}(3)$. Thus the triangle does not merely hold three phases: its one-particle edge sector carries the fundamental representation, its two-particle sector carries the dual representation, and its traceless bilinears carry the adjoint octet.

The triplet description of quarks and the three-quark construction of baryons originate with the quark model [19]; the additional color triplet was introduced to resolve the statistics and state-counting problem [20, 21]. The claim here is narrower and new: Tessera's three oriented edge modes would provide a geometric carrier of the same representation content, not a derivation of QCD from the combinatorics alone.

7.1 Geometric normalization

For the stored complex squared lengths $z_i = \rho_i e^{i\theta_i}$ on the three oriented edges, define

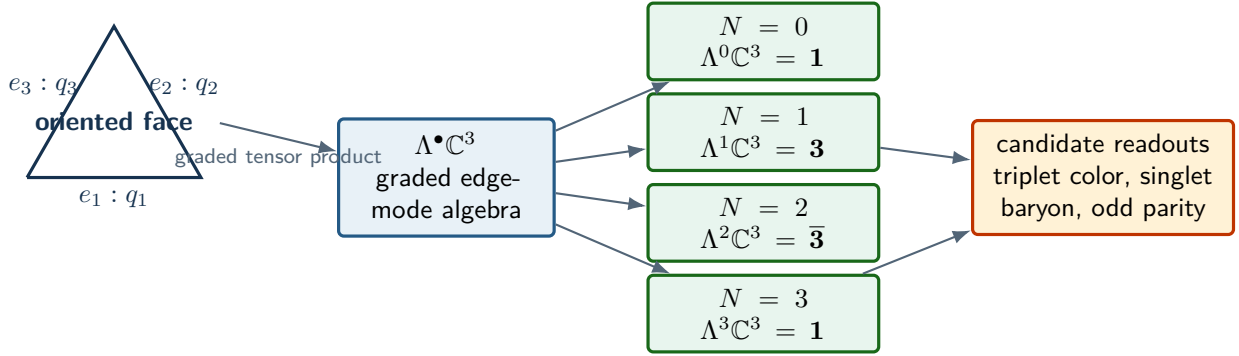


Figure 3: Exact representation content of three oriented edge-mode factors. The exterior sectors and their parity are algebraic identities. Interpreting the rank-three odd sector as quark color and the top wedge as a baryon color singlet is the physical hypothesis to be tested.

$$c_i = \frac{z_i}{\sqrt{|z_1|^2 + |z_2|^2 + |z_3|^2}}, \quad |c\rangle = \sum_{i=1}^3 c_i |i\rangle.$$

Then $\langle c|c\rangle = 1$. Constraining the perimeter to one is a valid geometric scale gauge, but it is an L^1 condition and does not replace the L^2 Hilbert normalization. Normalized pure color states form $\mathbb{CP}^2$; $SU(3)$ is the transformation group, not the surface of the triangle itself.

7.2 The existing omega phase pattern

Let $\omega = e^{2\pi i/3}$. The exact Fourier frame

$$F_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix}$$

is unitary. The existing pattern $(1, \omega, \omega^2)/\sqrt{3}$ is therefore one color basis vector, not by itself the whole color fiber. Its cyclic orbit supplies an exact orthonormal triad.

8 Color transport and Wilson loops without a new gauge field

Let T_{AB} be the chain-level transfer already induced by connecting simplices from component B to component A . In positive self-adjoint local spectral frames of common rank r , the raw fiber map is

$$M_{AB} = \Phi_A^\dagger W_A T_{AB} \Phi_B.$$

Its departure from an isometry is a physical leakage diagnostic:

$$\eta_{AB} = \|M_{AB}^\dagger M_{AB} - I\|.$$

When the selected band is isolated, M_{AB} has full numerical rank, and η_{AB} is small, take the polar unitary

$$V_{AB} = M_{AB}(M_{AB}^\dagger M_{AB})^{-1/2} \in U(r).$$

Under a change of local spectral frame, $\Phi_A \mapsto \Phi_A g_A$ and $\Phi_B \mapsto \Phi_B g_B$,

$$V_{AB} \mapsto g_A^\dagger V_{AB} g_B.$$

Consequently the full closed holonomy and its normalized trace transform by conjugation at the base point:

$$H(\gamma) = \prod_{(AB) \in \gamma} V_{AB}, \quad W_{U(r)}(\gamma) = \frac{1}{r} \text{Tr } H(\gamma).$$

At rank three there is no globally single-valued operation $V \mapsto V/(\det V)^{1/3}$: the cube root is $\mathbb{Z}_3$ -ambiguous. The faithful derived datum is therefore retained as

$$V_{AB} \in U(3), \quad \delta_{AB} = \det V_{AB} \in U(1), \quad [V_{AB}] \in PU(3) \cong SU(3)/\mathbb{Z}_3.$$

A local $SU(3)$ lift $\tilde{U}_{AB} = \delta_{AB}^{-1/3} V_{AB}$ may be followed continuously along a path after fixing a base branch, but its accumulated center sector must be reported. Fundamental Wilson traces use the full $U(3)$ holonomy or an explicitly lifted path; adjoint/projective Wilson loops are center-blind and require no branch. This turns the former ambiguity into two measured sectors rather than silently discarding one.

The determinant line also supplies a possible oriented flux readout. For a closed, full-rank world-tube family $V(t)$,

$$\nu = \frac{1}{2\pi} \oint d \arg \det V(t) \in \mathbb{Z}$$

is homotopy-invariant while the gap and rank remain open, and changes sign when the tube orientation is reversed. Identifying $B = \nu/3$ for an accepted quark tube is a proposed physical interpretation, not a group identity. Conservation under pair creation is exact only for a continuous conjugate-pair homotopy with no determinant zero or boundary flux.

For non-normal bands the correct overlap is biorthogonal,

$$M_{AB} = \Psi_A^\dagger W_A T_{AB} \Phi_B, \quad \Psi_C^\dagger W_C \Phi_C = I.$$

It is generally a $GL(r, \mathbb{C})$ transport, not a unitary one; left and right residuals, singular values, and frame condition numbers are part of the observable. In an indefinite Hermitian sector the Krein inertia is reported and a pseudo-unitary reduction is attempted only when the two signatures agree. No $U(r)$ or $SU(3)$ Wilson value is emitted by silently applying the positive-metric formula outside its domain.

Polar normalization must never conceal a bad fiber assignment: every accepted holonomy is reported together with η_{AB} , rank/singular value thresholds, endpoint band gaps, and frame conditioning.

This construction is the discrete spectral-frame analogue of Berry transport [22] and its non-Abelian Wilczek-Zee generalization [23]. Overlap matrices are also a standard route to gauge-covariant lattice observables [24]. Wilson loops themselves are foundational lattice gauge observables [25]. What is specific here is that the link matrix is not independently assigned: it is reconstructed from neighboring Hodge frames and accompanied by a leakage certificate.

9 Quarks as modular clusters

A *quark candidate* is proposed to be a component Q satisfying all of the following derived conditions:

The abstract rank-three band must first be anchored to oriented two-simplices. For an oriented triangle $\tau \subset Q$, let $R_\tau : C_1(Q) \rightarrow \mathbb{C}^3$ restrict a one-chain to the three ordered boundary edges, including their incidence signs. For a rank-three frame Φ_Q , define the gauge-invariant anchor score

$$a_Q^2 = \sum_{\tau \subset Q} w_\tau |\det(R_\tau \Phi_Q)|^2, \quad w_\tau \geq 0, \quad \sum_{\tau} w_\tau = 1.$$

The phases of the nonzero determinants give a local determinant-line trivialization; their coherence on overlapping triangles is recorded separately. Concentration on one triangle is a sufficient special case, not a requirement: an extended fiber may be anchored by an atlas of oriented faces. Because every 2-simplex has exactly three boundary edges, this anchor is independent of the ambient spectral dimension.

1. Q is a persistent high-modularity cluster, not a prescribed region.
2. Its selected color fiber has stable rank three.
3. Its triangle-anchor score and determinant-line coherence are stable.
4. It occupies an odd exterior sector.
5. Its color transport has bounded leakage over its lifetime.
6. Its determinant line has an accepted oriented winding $\nu = \pm 1$; reversing the tube yields the dual color representation and an antiquark.
7. Its total spectral fingerprint is stable under refinement and vertex relabeling.

The distinction between an antiquark and the $\Lambda^2 \mathbb{C}^3$ anti-triplet of two quarks is made by determinant-line orientation and total occupation, not color alone. Assigning $B = \nu/3$ is accepted only when the winding certificate above exists. A forward/reverse pair then has zero total winding under a gap-preserving conjugate homotopy; without that certificate baryon number remains unknown rather than being inserted by definition.

Flavor and electric charge are not assumed as hidden labels. The conservative hypothesis is that two stable subclasses of the same cluster fiber provide an isospin doublet. On such a doublet, the measured orientation flux supplies baryon number and the standard relation

$$Q = I_3 + \frac{B}{2}$$

gives $Q_u = +2/3$ and $Q_d = -1/3$. This is a proposed identification, not yet a derivation: it succeeds only if an unlabeled two-dimensional spectral band emerges, is transported coherently, and its Gauss-flux readout agrees with those values.

10 Fermion statistics from simplicial orientation

The graded interchange law is

$$\tau(a \hat{\otimes} b) = (-1)^{F_a F_b} b \hat{\otimes} a.$$

Two odd clusters therefore acquire a minus sign on exchange, while an even composite does not. Parity adds modulo two:

Object	Odd constituents	Composite parity
quark or antiquark	1	odd
meson $q\bar{q}$	2	even
diquark qq	2	even
baryon qqq	3	odd

Pauli exclusion is the exact exterior-algebra identity

$$\|v_1 \wedge \cdots \wedge v_n\|^2 = \det[\langle v_i, v_j \rangle].$$

If two complete one-particle modes coincide, the determinant and the state vanish. The “complete” qualifier prevents double-counting signs: color, spin, flavor, space, and component support are wedged once as one mode. One must not multiply an extra fermion sign by the sign already present in the color epsilon tensor.

10.1 A label-independent exchange experiment

Let Φ_t be an orthonormal frame for the isolated odd subspace at frame t . Define the parallel transport

$$R_t = \text{polar}(\Phi_{t+1}^\dagger W_t \Phi_t), \quad U_\gamma = R_{T-1} \cdots R_0.$$

The raw determinant line contains both permutation statistics and an ordinary Abelian Berry phase:

$$\chi_{\text{raw}}(\gamma) = \det U_\gamma.$$

It is therefore not expected to equal ± 1 on a generic geometric loop. Let γ_0 be a non-exchanging reference motion with the same geometric footprint, timing, and local frame convention. The interferometric exchange character is

$$\hat{\chi}_F(\gamma) = \frac{\det U_\gamma}{\det U_{\gamma_0}}.$$

The proposed dynamical test is

$$\hat{\chi}_F(\text{single exchange}) = -1, \quad \hat{\chi}_F(\text{double exchange}) = +1.$$

As an independent structural cross-check, persistent component matching extracts the permutation P_γ of localized odd blocks and reports $\text{sgn } P_\gamma$ together with the norm of the residual in-block motion after comparison with the reference loop. The algebraic wedge sign is exact; the interferometric holonomy is the dynamical certificate.

A physical 2π rotation uses the same reference normalization. If an emergent manifold-like regime supplies tangent frames, a continuum spinor claim additionally requires a lift of the frame holonomy from $SO(d)$ to $\text{Spin}(d)$; obstruction by the second Stiefel–Whitney class is then a falsification certificate [26]. This requirement concerns the physical spin lift, not the existence of the abstract CAR/Fock algebra. The spin-statistics comparison is cleanest as

$$\widehat{\chi}_F(\text{exchange})\widehat{\chi}_F(2\pi \text{ rotation})^{-1} = +1,$$

with each factor separately near -1 .

Configuration-space topology already explains how exchange classes can carry quantum phases [27] and, in two dimensions, more general statistics [28]. The Tessera proposal uses this precedent only as a diagnostic template. The minus sign from the graded exterior algebra is exact; the claim that an actual geometric exchange cobordism realizes the corresponding determinant holonomy remains an experiment.

11 Fock space as an inductive limit of interactions

For M oriented fermionic edge modes,

$$\widehat{\bigotimes}_{m=1}^M \mathbb{C}^2 \cong \Lambda^\bullet \mathbb{C}^M = \bigoplus_{n=0}^M \Lambda^n \mathbb{C}^M,$$

and the dimension identity is exact:

$$2^M = \sum_{n=0}^M \binom{M}{n}.$$

The exterior algebra is canonical as a functor of the one-particle space. Writing it as a literal ordered qubit tensor product, or implementing creation operators by Jordan–Wigner/bitset strings, requires a chosen mode order. Tessera derives a deterministic order from oriented component lineage and applies the parity of every reordering; all reported observables must be invariant under relabeling plus that induced unitary.

Adding a new noninteracting mode uses the vacuum embedding

$$\iota_M : \mathcal{H}_M \hookrightarrow \mathcal{H}_{M+1}, \quad \iota_M(\psi) = \psi \widehat{\otimes} |0\rangle.$$

The infinite Fock space is the direct limit

$$\mathcal{F} = \varinjlim (\mathcal{H}_M, \iota_M).$$

This makes the infinite expansion precise without ever allocating an infinite array. At every finite simulation time only finitely many modes have interacted. Consistency requires

$$\|\iota_M U_M - U_{M+1} \iota_M\| \longrightarrow 0$$

over a refinement sequence.

Bosonic gauge excitations need not add a new local oscillator. The traceless even bilinears

$$a_i^\dagger a_j - \frac{1}{3} \delta_{ij} N$$

span the color octet in $\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{1} \oplus \mathbf{8}$ and have even fermion parity. Arbitrarily many such collective excitations are represented by adding more microscopic modes at finer resolution. Each finite edge-mode factor remains two-dimensional.

This scale-by-scale state growth is adjacent to entanglement renormalization, where local Hilbert data are reorganized across layers before truncation [29]. The distinction is material: Tessera uses static/shifted response reduction plus an inductive vacuum embedding, and it must certify compatibility between successive finite spaces rather than assume a fixed bond dimension.

11.1 Occupation exterior algebra is not automatically Kähler–Dirac

The exterior algebra above is over the one-particle *mode space*; its degree is occupation number. A Kähler–Dirac field instead lives on the inhomogeneous differential-form/cochain space $\bigoplus_k C^k(K)$ and is acted on by $d - d^*$ (or a related Dirac square root). These constructions share exterior-algebra notation but are not the same operator or grading. Consequently the present model does not inherit lattice taste multiplicity merely from using $\Lambda^\bullet \mathfrak{h}_K$.

If a later Tessera model promotes its one-particle field to all cochain degrees and uses the Kähler–Dirac operator, the known flat four-dimensional decomposition into four Dirac spinors becomes an expected spectrum diagnostic, not an unexplained bug [30, 31]. Any observed near-fourfold cluster in the present model is reported as an empirical degeneracy until that stronger operator identification is made.

12 The proton as the maximally informative baryon

The proton is chosen because, beyond generic baryon structure, it demands a nontrivial flavor pattern, electric charge, spin, and experimentally meaningful form factors.

Let three persistent quark components A, B, C have color frames and normalized color columns c_A, c_B, c_C . The invariant color volume is

$$S_{ABC} = \epsilon_{ijk} c_A^i c_B^j c_C^k = \det[c_A \ c_B \ c_C].$$

Under a common $g \in SU(3)$, $S \mapsto \det(g)S = S$. Its squared magnitude is the Gram determinant

$$|S_{ABC}|^2 = \det(C^\dagger C) \in [0, 1].$$

The value one means the three color directions form an orthonormal frame. Their normalized wedge is then the unique $\Lambda^3 \mathbb{C}^3$ singlet. The proposed proton certificate is the conjunction:

- three persistent odd rank-three quark clusters with accepted triangle-anchor certificates;
- one persistent bound supercluster containing them;
- normalized color wedge with $|S_{ABC}|^2 \approx 1$ and vanishing net color flux;
- flavor spectrum with the uud occupation pattern;
- oriented baryon flux $B = 1$;
- Gauss flux $Q = +1$;
- total-space spin holonomy $J^2 = 3/4$, a reference-normalized $2\pi \mapsto -1$, and, where applicable, an accepted spin lift;
- finite radius and stable spectral mass/form-factor readouts; and
- stability of every dimensionless certificate under refinement.

None of these conditions should be included as an emergence target. The proton is found only if the base geometric optimization produces a component satisfying them. Targeted runs remain valuable as existence and obstruction experiments, but must be labeled as synthesis rather than emergence.

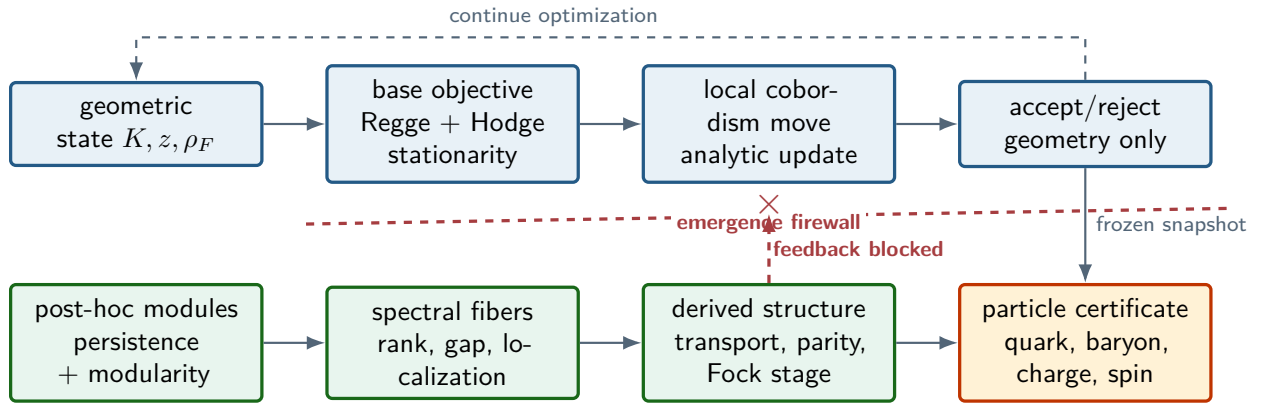


Figure 4: No-feedback emergence protocol. Only the base geometric objective drives optimization. Cluster, fiber, color, exchange, and baryon observables are computed from accepted snapshots and cannot feed back into the objective in an emergence run. Targeted synthesis is a separate, explicitly labeled mode.

13 The master recursive construction

Let $\mathcal{R}_0(\lambda) = L_0 - \lambda I$ denote the microscopic one-particle response pencil. At every scale:

$$\begin{aligned}
 P_\ell &= \text{PersistentPartition}(\mathcal{R}_\ell), \\
 E_v^{\ell+1} &= \text{certified isolated subspace of } C_v^\ell, \\
 \mathcal{R}_{\ell+1}(\lambda) &= \text{Feshbach}_{P_\ell}(\mathcal{R}_\ell(\lambda)), \\
 V_{vw}^{\ell+1} &= \text{Polar}_{r_v}((\Phi_v^\ell)^\dagger W T_{vw} \Phi_w^\ell), \\
 \mathcal{H}_{\ell+1} &= \mathcal{F}_- \left(\bigoplus_v E_v^{\ell+1} \right).
 \end{aligned}$$

At $\lambda = 0$ the response step is the exact supported static Schur complement. For a nonzero band it

is the exact energy-dependent pencil; a cached linear $\mathcal{R}_{\ell+1}$ is an AMLS/component-mode surrogate with a declared frequency window and residual. The transport rank is generic; only an anchored accepted rank-three fiber receives the color interpretation. This recursion supplies the response network, retained stalk, derived transport, and expanding state space without claiming that every coarse level is literally a new simplicial complex.

14 Exactness and performance principles

The simulation should prefer an exact structural identity over a general dense numerical operation whenever both compute the same object:

- use sparse static and shifted Schur/Feshbach solves, not explicit dense inverses, and use AMLS when a reusable linear band surrogate is needed;
- use Künneth sums only for actual product complexes, and occupation subset sums for $d\Gamma(L)$, not diagonalization of an eager Fock matrix;
- use exterior bit parity for exchange signs, not sampled phases;
- use exact 3×3 determinants and the fixed F_3 frame only after a rank-three band passes its triangle-anchor certificate;
- use analytic Regge/Hodge derivatives and Wirtinger gradients, not finite differences;
- use Smith normal form for integer homology and a spectral threshold only as a cross-check;
- use matrix-determinant/Woodbury updates for local cobordism changes;
- cache component factorizations and invalidate only affected stars;
- keep tensor products lazy and block-sparse by occupation/parity;
- use $U(r)$ polar transport at generic rank, retaining determinant-line and projective/center data at $r = 3$; and
- attach frequency window, residual, gap, leakage, signature, and condition-number certificates to every iterative eigensolve or reduction.

The exact route is not only faster. It prevents a numerical tolerance from becoming an undocumented physical postulate.

15 Prior art and boundary of novelty

No single cited work establishes the full recursive spectral-fiber proposal. The construction is a synthesis of several mature ideas, and its novelty should be evaluated at the joins. Table 3 states the boundary explicitly.

Table 3: Established ingredients and the additional Tessera claim.

Topic	Established prior art	Additional claim made here
Simplicial geometry and Hodge spectra	Regge curvature, discrete exterior calculus, and combinatorial Laplace spectra [2–4].	Use one jointly optimized Regge-Hodge complex as the only microscopic carrier for geometry and quantum readouts.

Topic	Established prior art	Additional claim made here
Coarse response and recursive modules	Kron/Schur reduction, Feshbach maps, component-mode synthesis/AMLS, cellular sheaf Laplacians, modular communities, and self-similar network renormalization [5–8, 10–13].	Treat a persistent component as a static response vertex; retain nonzero bands through shifted or certified component-mode reduction; recurse in operator-valued response networks, using a sheaf realization only when its factorization is certified.
Geometric gauge transport	Berry and Wilczek-Zee holonomy, overlap-based lattice links, connection Laplacians/vector diffusion, and Wilson loops [22–25, 32].	Derive $U(r)$ transport from component frames; at anchored rank three retain the determinant line, projective $SU(3)/\mathbb{Z}_3$ class, and any chosen center lift, assigning no independent gauge link variable.
Color and fermion structure	Quark/color triplets, exterior Fock/second quantization, topological exchange phases, and spin structures [14, 19–21, 26–28].	Realize $\mathbf{1} \oplus \mathbf{3} \oplus \bar{\mathbf{3}} \oplus \mathbf{1}$ on three oriented edge modes, anchor abstract rank-three fibers to oriented faces, and test exchange by a Berry-cancelled determinant-line interferometer plus structural permutation parity.
Scale composition and boundaries	TQFT cobordisms, general-boundary state assignments, categorical tensor composition, second quantization, and entanglement renormalization [14–17, 29].	Keep simplicial gluing at the one-particle level, then build finite Fock stages functorially and require vacuum-embedding compatibility under refinement.
Kähler–Dirac boundary	Differential-form fermions and their taste structure [30, 31].	Do not infer Kähler–Dirac tastes from occupation exterior algebra; test for them only if the one-particle field is promoted to inhomogeneous cochains with a Kähler–Dirac operator.
Spectral spacetime	Diffusion spectral dimension on ensembles of simplicial geometries [1].	Test whether many interacting Tessera cobordisms yield a stable four-dimensional spectral window while simultaneously supporting the particle certificates.

16 Falsification program

The formulation fails, or must be narrowed, if any of the following persists under refinement and tighter numerical certification:

1. **No persistent rank-three clusters.** High-modularity components appear, but their localized fiber rank or spectral gap is unstable.
2. **No oriented color anchor.** A rank-three band appears, but its projected alternating volume on oriented triangles vanishes or drifts.
3. **No faithful coarse response.** Schur-reduced components fail to reproduce static response, or shifted/AMLS reduction fails over its declared frequency band, within the stated residual.
4. **No derived gauge covariance.** Wilson values depend on the local spectral frame after leakage is controlled, or the determinant and $\mathbb{Z}_3$ center sectors cannot be made path-consistent.

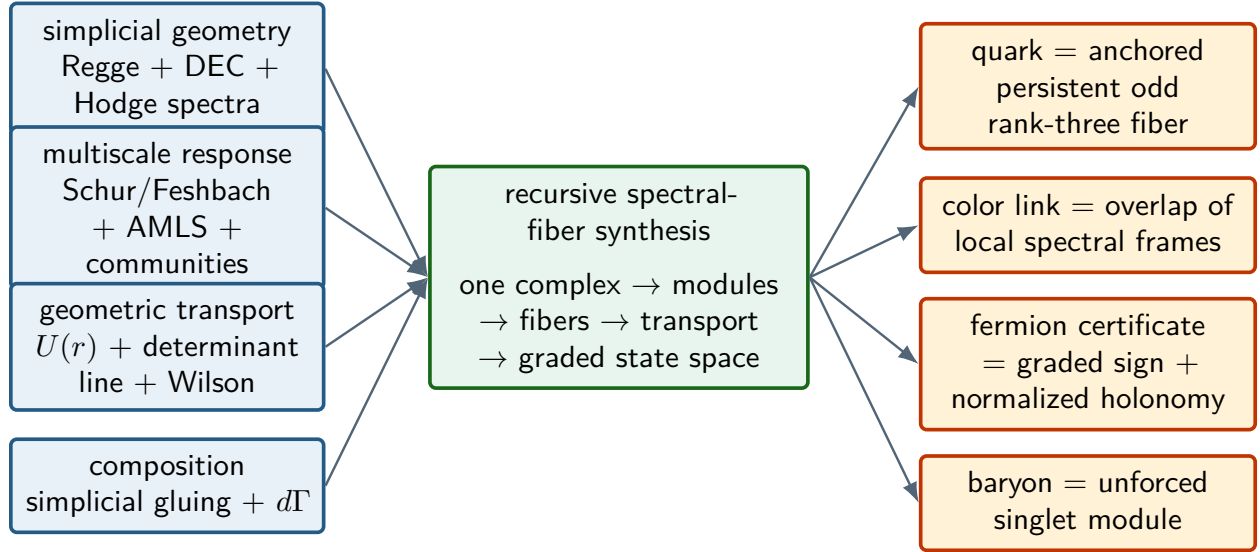


Figure 5: Relationship to prior art. The blue inputs are established research programs; the green center is the proposed Tessera synthesis; the amber outputs are new physical identifications and must be validated independently. An arrow denotes conceptual inheritance, not a proof of the downstream claim.

5. **No fermion holonomy.** The Berry-cancelled exchange ratio or structural permutation sign does not give -1 , or the verdict changes under relabeling.
6. **No spinor rotation.** Exchange works but the reference-normalized 2π physical rotation does not give -1 ; in a manifold-like continuum claim, failure of a consistent spin lift is also decisive.
7. **No inductive compatibility.** Adding vacuum modes changes already-computed amplitudes by a nonvanishing amount.
8. **No unforced baryon.** Targeted synthesis can build the certificates, but the stationary geometric ensemble never produces them without a proton-specific term.
9. **No continuum stability.** Dimensionless color, parity, charge, spin, and amplitude certificates drift rather than converge with refinement.
10. **Unexpected multiplicity.** A robust flavor/taste degeneracy is neither predicted by the stated one-particle operator nor stable enough to be promoted to an emergent flavor mechanism.

Holes may re-emerge and may correlate with some phases, but no claim in this paper depends on them doing so.

17 Conclusion

The revised geometry is economical but more precise: an edge carries a two-level mode, not an independently stored pure state; a quark candidate is a modular spectral component whose rank-three band is anchored to oriented faces; its transport is a certified $U(3)$ overlap with retained determinant-line and projective color data; and its fermionic sign is the exterior grading, checked dynamically only after cancelling ordinary Berry phase. Simplicial gluing constructs the one-particle operator and second quantization constructs the expanding Fock state. Three accepted components form a baryon through their normalized color wedge; the proton is the sharpest test because it also demands the correct determinant-line flux, charge, flavor, and reference-normalized spin response.

The exact claims are now limited to their proper domains: static Schur response, energy-dependent Feshbach isospectrality, exterior/CAR algebra, second-quantized direct-sum composition, and gauge covariance of accepted transport. What remains genuinely open is whether Tessera’s unforced Regge–Hodge dynamics produces the required anchored clusters, low-leakage holonomies, determinant windings, and spin lift. A tempting Hellmann–Feynman/envelope argument does not by itself make the first variation of transport Gram defect vanish at a Regge–Hodge stationary point, because the defect is not the optimized functional. Tessera will therefore measure that correlation as a conjectural scaling law rather than cite stationarity as a theorem.

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Repository evidence

- Tessera amplitude and obstruction results, [cobordism-results.md](#).
- Tessera spectral-dimension status, [h_ds4_status.md](#).
- Tessera interaction-history construction, [interaction-history-monte-carlo.md](#).
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- Existing fixed-partition modularity implementation, [ModularityOptimizer.h](#).
- Current visualization and joint-stationarity experiment, [proton_animation.py](#).
- External-review disposition and exact-claim ledger, [referee response](#).